

Brief summary of the present ^{273}Ds and ^{269}Hs analysis

Z. R. Sun

Purpose

This note summarizes the current experimental evidence and tentative structural interpretation of two decay chains assigned to ^{273}Ds , observed in the $^{238}\text{U}(^{40}\text{Ar}, 5n)^{273}\text{Ds}$ reaction at the SHANS-2 separator. The manuscript is still under preparation, and the state assignments discussed below should be regarded as working hypotheses rather than final level assignments.

The main point on which we would appreciate expert advice is whether the lower-energy, longer-lived ^{273}Ds decay observed in the present work should be connected with the corresponding long-lived event reported in the DGFRS-2 experiment, or whether the two events should be kept separate at the present stage. This choice affects the estimated hindrance factor and the possible Nilsson-configuration assignment.

Experimental results

The experiment was performed at SHANS-2 using the $^{238}\text{U}(^{40}\text{Ar}, xn)^{278-x}\text{Ds}$ reaction. At an average excitation energy of about 50 MeV, two complete decay chains assigned to the $5n$ channel were observed. The event-by-event decay sequences are shown in Fig. 1, and the corresponding decay properties are summarized together with previously reported chains in Table 1. The subsequent decays in both observed chains proceed through ^{269}Hs , ^{265}Sg , and terminal spontaneous fission of ^{261}Rf , and are consistent with previously reported decay properties of these nuclides [1].

The first chain starts with a higher-energy and shorter-lived ^{273}Ds alpha decay, whereas the second chain starts with a lower-energy and longer-lived decay. This two-branch pattern resembles that reported in the DGFRS-2 $^{40}\text{Ar} + ^{238}\text{U}$ experiment [2], but the present chains are complete and therefore provide additional information on the daughter decays.

Relation to the DGFRS-2 result

The long-lived ^{273}Ds alpha decay in the present work has

$$E_\alpha = 10.858(22) \text{ MeV}, \quad \Delta t = 8.320 \text{ ms.}$$

The corresponding long-lived event reported in the DGFRS-2 experiment [2] has

$$E_\alpha = 10.929(22) \text{ MeV}, \quad \Delta t = 41.703 \text{ ms.}$$

The energy difference is about 71 keV, which is larger than three times the present experimental energy uncertainty. Therefore, although both events belong to the lower-energy, longer-lived branch relative to the known high-energy ^{273}Ds decay, it is not obvious whether they should be assigned to the same state. The comparison of the available E_α and decay-time information in Fig. 2 illustrates this ambiguity, and we are considering both possibilities.

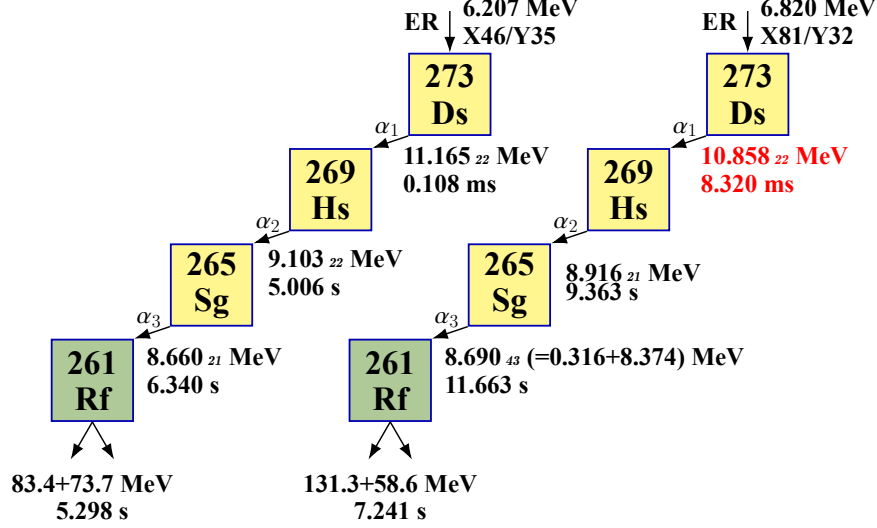


Figure 1: Two decay chains assigned to ^{273}Ds observed in the $^{238}\text{U}(^{40}\text{Ar}, 5n)^{273}\text{Ds}$ reaction. For each chain, the energy of the evaporation residue (ER) event and its X/Y position on the detector are given at the top, followed by the energies of the successive α -decay events and the terminal spontaneous-fission (SF) event, together with the time intervals between adjacent events. Energies in parentheses denote summed signals. Italic values indicate the corresponding energy uncertainties. Red text validated in the present work and the inferred assignments related to the isomeric states.

Possible low-lying state in ^{269}Hs

The daughter decay in the long-lived ^{273}Ds chain has $E_\alpha = 8.916$ MeV, which is lower than the higher-energy ^{269}Hs alpha group near 9.0–9.3 MeV. In the present analysis, the lower-energy ^{269}Hs group is preferentially associated with the lower-energy, longer-lived ^{273}Ds branch, whereas the higher-energy ^{269}Hs group is mainly associated with the higher-energy, shorter-lived ^{273}Ds branch. No clear cross feeding has been observed within the available statistics.

Table 1: Decay properties of the observed chains assigned to ^{277}Cn , ^{273}Ds , ^{269}Hs , ^{265}Sg , and ^{261}Rf . The decay times of individual events are listed.

No.	Reference	^{277}Cn		^{273}Ds		^{269}Hs		^{265}Sg		^{261}Rf	
		E_α	Δt (ms)	E_α	Δt (ms)	E_α	Δt (s)	E_α	Δt (s)	E_α	Δt (s)
Ch1	[3]	11.45	0.280	11.08	0.110	9.23	19.7	(4.60)	7.4	8.52	4.7
Ch2	[3]	11.17	1.406	11.20	0.310	9.18	22.0	(0.2)	18.8	(SF)	14.5
Ch3	[4]	11.09	1.10	11.14	0.520	9.17	14.2	8.71	23.0	(SF)	2.97
Ch4	[4]	11.32	1.22	11.15	0.0399	9.25	0.270	8.70	79.0	(SF)	8.30
Ch5	[5]	11.07	0.370	11.03	0.373	9.15	36.0	8.66	13.8	(SF)	3.73
Ch6	[2]	–	–	11.017	0.184	–	–	8.397	9.0766	(SF)	3.0344
Ch7	[2]	–	–	10.929	41.703	(0.342)	4.0978	8.509	58.19	(SF)	0.3923
Ch8	This work	–	–	11.165	0.108	9.103	5.006	8.660	6.340	(SF)	5.298
Ch9	This work	–	–	10.858	8.320	8.916	9.363	8.690	11.663	(SF)	7.241

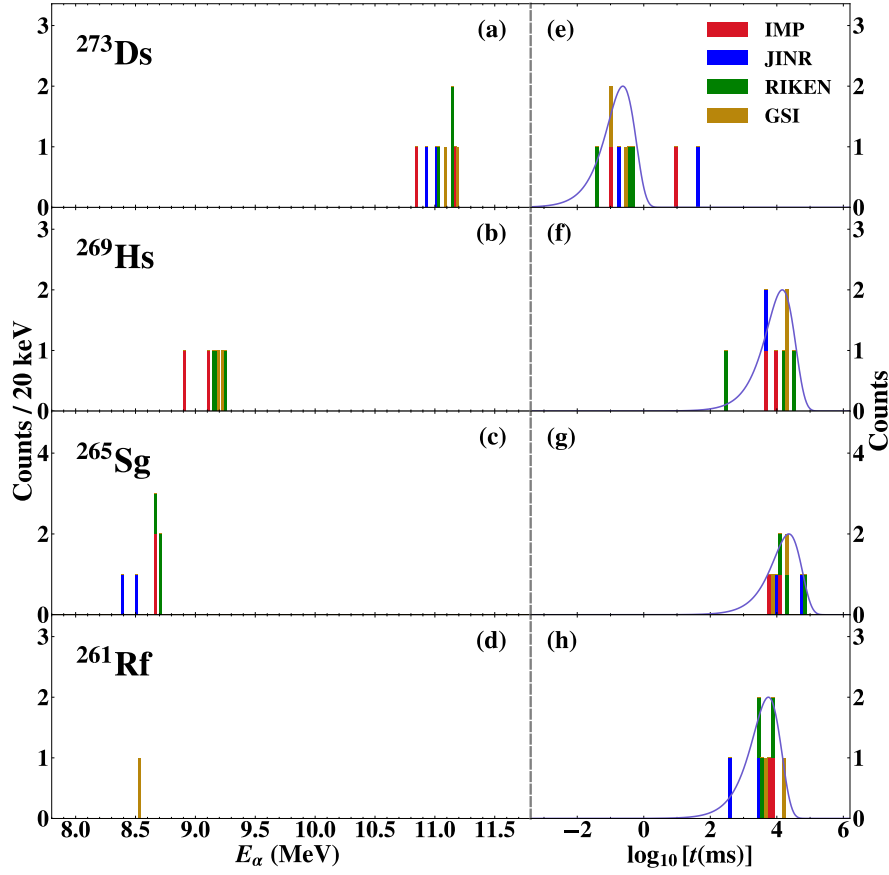


Figure 2: Comparison of the E_α and mean life information for decay chains involving ^{273}Ds . The red, blue, green, and yellow symbols correspond to the IMP, JINR [2], RIKEN [4, 5], and GSI [3, 6]. The fit curves in the decay-time panels are drawn according to Ref. [7].

This apparent selectivity in the α -decay feeding makes it more plausible that the 8.92–9.00 MeV group corresponds to an additional low-lying or isomeric state in ^{269}Hs , rather than merely to an unresolved α -decay branch from the same state. The compiled ^{269}Hs α -energy distribution in Fig. 3 shows how the present low-energy event compares with the directly produced and daughter-populated ^{269}Hs data. Owing to the inability of previous radiochemical experiments to determine the ER– α_1 correlation time, the present work provides the first direct lifetime measurement for this low-energy α -decay group in ^{269}Hs . However, its measured lifetime does not appear to differ significantly from those of the other groups. Given the still limited number of events, the proposed classification of states in ^{269}Hs should therefore remain tentative.

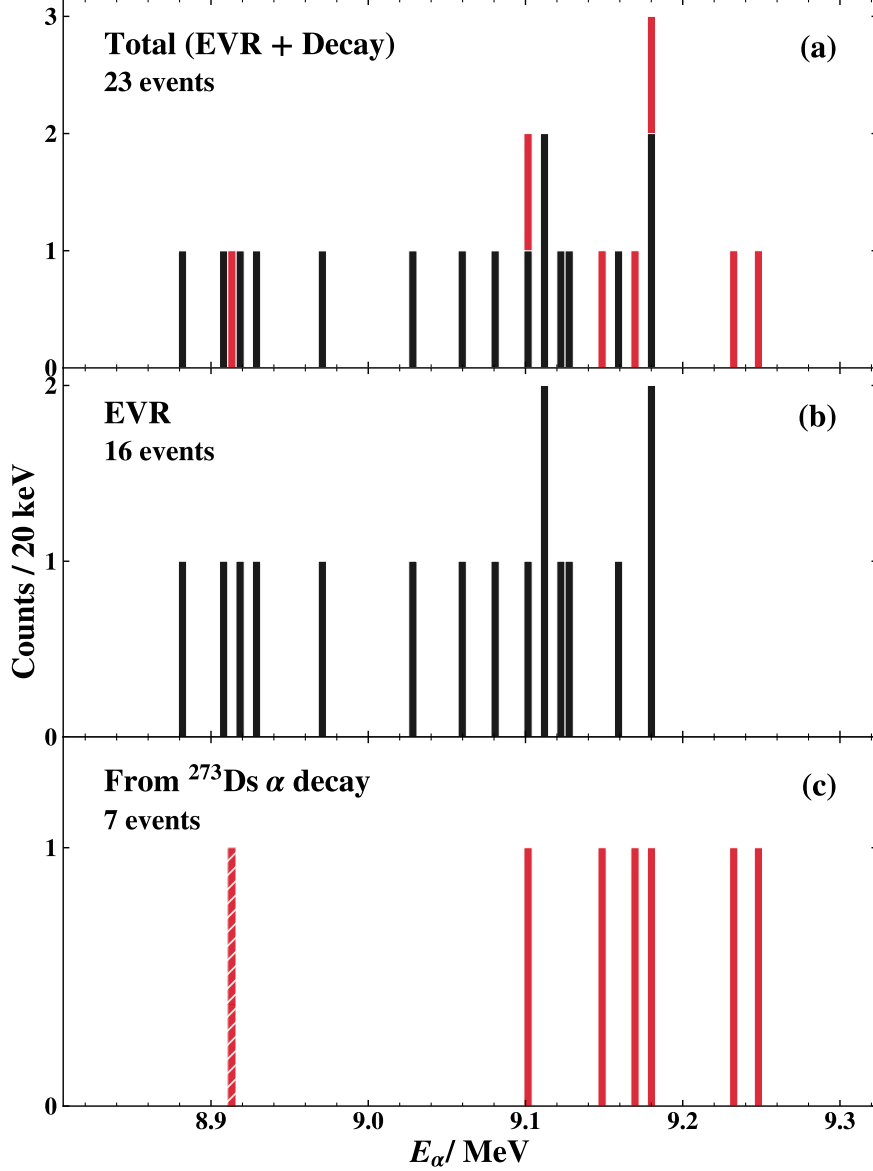


Figure 3: Energy distributions of α particles assigned to ^{269}Hs . The upper panel shows all events, the middle panel shows nuclei produced directly as evaporation residues (EVRs) [8–11], and the lower panel shows nuclei populated through the α decay of ^{273}Ds [3–5]. The hatched bin indicates the event observed in the present work at $E_\alpha = 8.915$ MeV.

Tentative structural interpretation

The structural interpretation is based on configuration-constrained potential-energy-surface calculations in the deformation space $(\beta_2, \beta_4, \beta_6)$. The nuclear surface is parameterized in terms of spherical harmonics, and the total energy is calculated within the macroscopic–microscopic Strutinsky framework using a realistic phenomenological mean-field Hamiltonian [15–17]. The total energy combines a macroscopic contribution from the standard liquid-drop model [18] with a microscopic contribution comprising the Strutinsky shell correction [19] and the pairing correlation energy. The single-particle spectrum is generated from a deformed Woods–Saxon potential using the universal parameter set [20], while pairing correlations are treated with the approximate particle-number-conserving Lipkin–Nogami method [21], adopting monopole

Table 2: Decay properties assigned to the long- and short-lived states of ^{273}Ds . The half-life of $^{273}\text{Ds}^a$ was obtained from a combined analysis of the long-lived decay events observed at JINR and IMP, whereas $^{273}\text{Ds}^b$ denotes the short-lived component. The value for $^{273}\text{Ds}^c$ was deduced from the single long-lived event observed at IMP only.

Nucleus	$T_{1/2}$ (ms)	E_α (MeV)	HF [12, 13]	HF [14]
$^{273}\text{Ds}^a$	$17.34^{+31.55}_{-6.83}$	10.89(2)	97^{+183}_{40}	101^{+192}_{42}
$^{273}\text{Ds}^b$	$0.16^{+0.08}_{-0.05}$	11.17(2)	$3.8^{+1.9}_{-1.2}$	$4.2^{+2.1}_{-1.3}$
$^{273}\text{Ds}^c$	$5.76^{+27.62}_{-2.64}$	10.86(2)	27^{+130}_{13}	28^{+135}_{-13}

pairing strengths G determined by the average-gap prescription [22]. For each specified one-quasineutron configuration, the unpaired neutron orbital is tracked and blocked at every point on the deformation lattice, thereby treating deformation and pairing consistently for that configuration. The hindrance factors used to distinguish the long- and short-lived ^{273}Ds components are listed in Table 2, while the calculated low-lying one-quasineutron configurations in ^{273}Ds and ^{269}Hs are summarized in Table 3. Their comparison with the observed α -decay branches is used to examine possible Nilsson-configuration assignments around the deformed $N = 162$ shell closure.

Table 3: Calculated one-quasineutron configurations and excitation energies for ^{273}Ds and ^{269}Hs .

^{273}Ds		^{269}Hs	
$(\beta_2, \beta_4, \beta_6) = (0.215, -0.076, -0.006)$		$(\beta_2, \beta_4, \beta_6) = (0.231, -0.062, -0.012)$	
Configuration	E_x (MeV)	Configuration	E_x (MeV)
$3d_{5/2} \uparrow$: $3/2^+[622]$	0.7387	$1i_{11/2} \downarrow$: $7/2^+[613]$	0.0796
$1i_{11/2} \downarrow$: $9/2^+[604]$	0.6655	$1j_{15/2} \downarrow$: $11/2^- [725]$	0.0397
$3d_{5/2} \uparrow$: $5/2^+[613]$	0.4730	$2g_{7/2} \downarrow$: $1/2^+[620]$	0.0292
$2g_{7/2} \downarrow$: $3/2^+[611]$	0.3192	$2g_{9/2} \uparrow$: $9/2^+[615]$	0.0058
$1j_{15/2} \downarrow$: $13/2^- [716]$	0	$3d_{5/2} \uparrow$: $3/2^+[622]$	0

In this region, ^{273}Ds can be viewed as containing one neutron above the $N = 162$ gap, while ^{269}Hs contains one neutron hole below it. The long-lived ^{273}Ds branch may therefore reflect a high- Ω one-quasineutron configuration whose alpha decay is hindered by configuration mismatch and, possibly, by angular-momentum and parity changes.

Two working scenarios are being considered, as illustrated in Fig. 4.

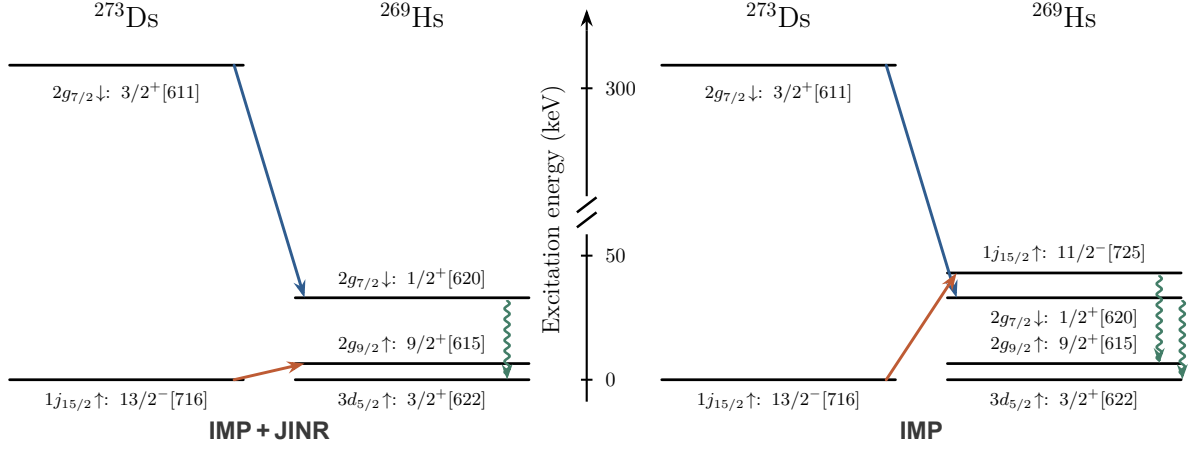


Figure 4: Comparison of the level assignments for the observed α -decay events. In the IMP+JINR interpretation (left), the two events are assigned to a common state in ^{269}Hs . In the IMP interpretation (right), the IMP event is assigned to a separate state in ^{269}Hs , followed by the indicated γ transition to the lower-lying state. The blue arrows denote the short-lived decay transitions, whereas the red arrows denote the long-lived decay transitions.

At present, neither the hindrance-factor analysis nor the configuration-constrained calculation provides a unique assignment. They are used mainly as consistency checks for possible structural interpretations.

Systematics and limitations

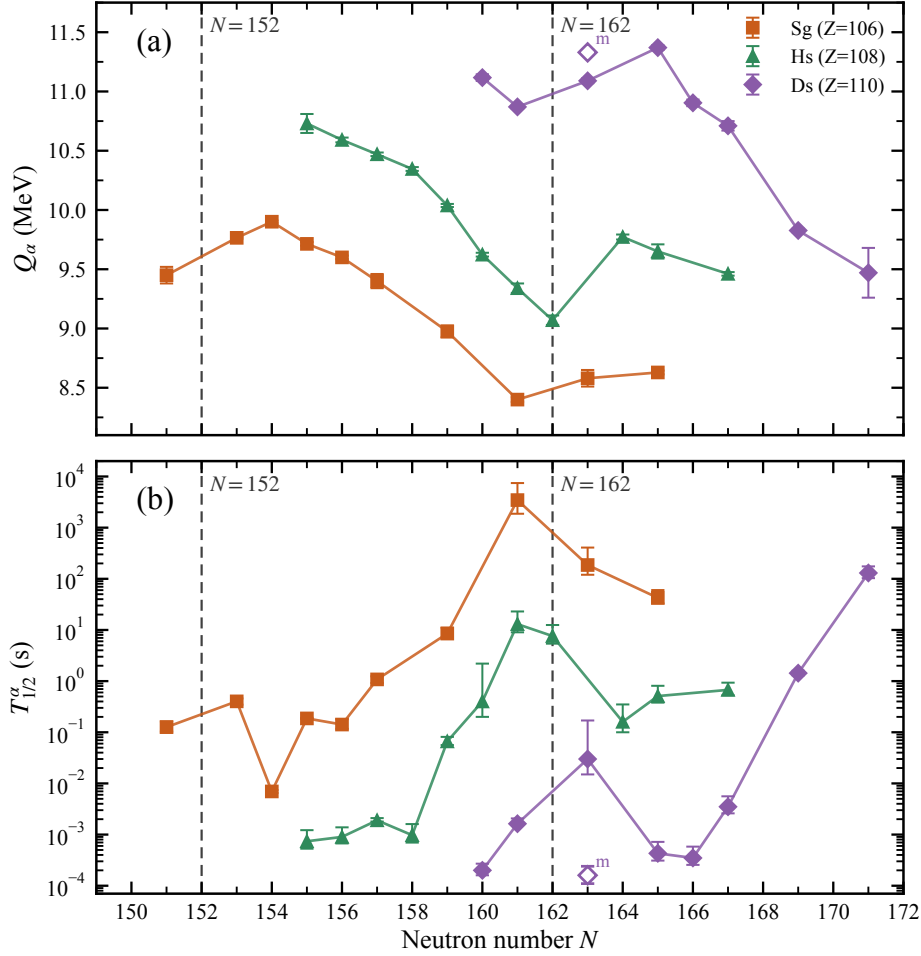


Figure 5: Systematics of Q_α values and partial α -decay half-lives for the Sg, Hs, and Ds isotopic chains. The dashed vertical lines mark $N = 152$ and $N = 162$. The open diamond labeled m represents $^{273}\text{Ds}^a$, whereas the filled diamond connected to the Ds isotopic trend represents $^{273}\text{Ds}^c$.

The isotopic systematics provide a useful broader context for the tentative assignments discussed above. As shown in Fig. 5, the Q_α values and partial α -decay half-lives in the Sg, Hs, and Ds chains do not evolve as a simple monotonic sequence across the deformed $N = 162$ region. The local behavior of the Hs and Ds chains is therefore qualitatively compatible with the possibility that different low-lying configurations contribute to the observed decay branches. In particular, the separation between the lower-energy, longer-lived ^{273}Ds branch and the higher-energy, shorter-lived branch is not inconsistent with a configuration-dependent α decay picture near the deformed shell closure.

This argument should remain modest. The same region is experimentally constrained by Extremely low fusion-evaporation cross sections, so several key nuclei are represented by only one or a few correlated decay chains, and some historical chains are incomplete. As a result, adopted half-lives, partial α -decay half-lives, and even branch assignments may be affected by low statistics, unresolved feeding. The systematics in Fig. 5 should therefore be treated as a consistency check that supports the plausibility of the proposed picture, not as an independent proof of a particular level ordering or Nilsson-configuration assignment. More complete, higher-statistics

decay data are needed to fix the ordering in ^{273}Ds and ^{269}Hs .

Open points for discussion

We would be grateful for comments on the following points.

1. Is the interpretation of the theoretical calculations presented in *Tentative structural interpretation* correct?
2. Are the following configuration assignments for the α -decay transitions physically reasonable,

$$^{273}\text{Ds}(\nu 13/2^- [716]) \xrightarrow{\alpha} ^{269}\text{Hs}(\nu 9/2^+ [615]), \quad \text{HF} \sim 100$$

$$^{273}\text{Ds}(\nu 13/2^- [716]) \xrightarrow{\alpha} ^{269}\text{Hs}(\nu 11/2^- [725]) \xrightarrow{\gamma} ^{269}\text{Hs}(\nu 9/2^+ [615]), \quad \text{HF} \sim 30$$

and

$$^{273}\text{Ds}(\nu 3/2^+ [611]) \xrightarrow{\alpha} ^{269}\text{Hs}(\nu 1/2^+ [620]) \xrightarrow{\gamma} ^{269}\text{Hs}(\nu 3/2^+ [622]), \quad \text{HF} \sim 4$$

or should alternative configurations be considered?

3. Based solely on the theoretical calculations, the following decay paths appear to be more plausible:

$$^{273}\text{Ds}(\nu 13/2^- [716]) \xrightarrow{\alpha} ^{269}\text{Hs}(\nu 11/2^- [725]) \xrightarrow{\gamma} ^{269}\text{Hs}(\nu 9/2^+ [615]), \quad \text{HF} \sim 30,$$

and

$$^{273}\text{Ds}(\nu 3/2^+ [611]) \xrightarrow{\alpha} ^{269}\text{Hs}(\nu 1/2^+ [620]) \xrightarrow{\gamma} ^{269}\text{Hs}(\nu 3/2^+ [622]), \quad \text{HF} \sim 4.$$

However, both proposed α -decay branches connect states dominated by one-quasineutron Nilsson configurations with $\Delta K = 1$. In addition, the Nilsson orbitals involved in each branch are interpreted as having closely related spherical parentage ($1j_{15/2}$ and $2g_{7/2}$). Why the two branches exhibit different inferred hindrance factors, $\text{HF} \sim 30$ and $\text{HF} \sim 4$?

References

- [1] H. Haba, D. Kaji, Y. Kudou, K. Morimoto, K. Morita, K. Ozeki, R. Sakai, T. Sumita, A. Yoneda, Y. Kasamatsu, Y. Komori, A. Shinohara, H. Kikunaga, H. Kudo, K. Nishio, K. Ooe, N. Sato, and K. Tsukada. Production of ^{265}Sg in the $^{248}\text{Cm}(^{22}\text{Ne}, 5n)^{265}\text{Sg}$ reaction and decay properties of two isomeric states in ^{265}Sg . *Physical Review C*, 85(2):024611, February 2012.
- [2] Yu. Ts. Oganessian, V. K. Utyonkov, M. V. Shumeiko, F. Sh. Abdullin, G. G. Adamian, S. N. Dmitriev, D. Ibadullayev, M. G. Itkis, N. D. Kovrizhnykh, D. A. Kuznetsov, O. V. Petrushkin, A. V. Podshibiakin, A. N. Polyakov, A. G. Popeko, I. S. Rogov, R. N. Sagaidak, L. Schlattauer, V. D. Shubin, D. I. Solovyev, Yu. S. Tsyganov, A. A. Voinov, V. G. Subbotin, N. S. Bublikova, M. G. Voronyuk, A. V. Sabelnikov, A. Yu. Bodrov, N. V. Aksenov, A. V. Khalkin, Z. G. Gan, Z. Y. Zhang, M. H. Huang, and H. B. Yang. Synthesis and decay properties of isotopes of element 110: ^{273}Ds and ^{275}Ds . *Physical Review C*, 109(5):054307, May 2024.

- [3] S. Hofmann, F.P. Heßberger, D. Ackermann, G. Münzenberg, S. Antalic, P. Cagarda, B. Kindler, J. Kojouharova, M. Leino, B. Lommel, R. Mann, A.G. Popeko, S. Reshitko, S. Šaro, J. Uusitalo, and A.V. Yeremin. New results on elements 111 and 112. *The European Physical Journal A*, 14(2):147–157, June 2002.
- [4] Kosuke Morita, Kouji Morimoto, Daiya Kaji, Takahiro Akiyama, Sin-ichi Goto, Hiromitsu Haba, Eiji Ideguchi, Kenji Katori, Hiroyuki Koura, Hisaaki Kudo, Tetsuya Ohnishi, Akira Ozawa, Toshimi Suda, Keisuke Sueki, Fuyuki Tokanai, Takayuki Yamaguchi, Akira Yoneda, and Atsushi Yoshida. Experiment on synthesis of an isotope $^{277}\text{112}$ by the $^{208}\text{Pb} + ^{70}\text{Zn}$ reaction. *Journal of the Physical Society of Japan*, 76(4):043201, April 2007.
- [5] Takayuki Sumita, Kouji Morimoto, Daiya Kaji, Hiromitsu Haba, Kazutaka Ozeki, Ryutaro Sakai, Akira Yoneda, Atsushi Yoshida, Hiroo Hasebe, Kenji Katori, Nozomi Sato, Yasuo Wakabayashi, Shin-ichi Mitsuoka, Shin-ichi Goto, Masashi Murakami, Yoshiki Kariya, Fuyuki Tokanai, Keita Mayama, Mirei Takeyama, Toru Moriya, Eiji Ideguchi, Takayuki Yamaguchi, Hidetoshi Kikunaga, Junsei Chiba, and Kosuke Morita. New result on the production of ^{277}Cn by the $^{208}\text{Pb} + ^{70}\text{Zn}$ reaction. *Journal of the Physical Society of Japan*, 82(2):024202, February 2013.
- [6] S. Hofmann, V. Ninov, F. P. Heßberger, P. Armbruster, H. Folger, G. Münzenberg, H. J. Schött, A. G. Popeko, A. V. Yeremin, S. Saro, R. Janik, and M. Leino. The new element 112. *Zeitschrift für Physik A Hadrons and Nuclei*, 354(3):229–230, December 1996.
- [7] K. H. Schmidt. A new test for random events of an exponential distribution. *The European Physical Journal A*, 8(1):141–145, July 2000.
- [8] J. Dvorak, W. Bröchle, M. Chelnokov, R. Dressler, Ch. E. Düllmann, K. Eberhardt, V. Gorschkov, E. Jäger, R. Krücken, A. Kuznetsov, Y. Nagame, F. Nebel, Z. Novackova, Z. Qin, M. Schädel, B. Schausten, E. Schimpf, A. Semchenkov, P. Thörle, A. Türler, M. Wegrzecki, B. Wierczinski, A. Yakushev, and A. Yeremin. Doubly magic nucleus $^{270}_{108}\text{Hs}_{162}$. *Physical Review Letters*, 97(24):242501, December 2006.
- [9] J. Dvorak, W. Bröchle, M. Chelnokov, Ch. E. Düllmann, Z. Dvorakova, K. Eberhardt, E. Jäger, R. Krücken, A. Kuznetsov, Y. Nagame, F. Nebel, K. Nishio, R. Perego, Z. Qin, M. Schädel, B. Schausten, E. Schimpf, R. Schuber, A. Semchenkov, P. Thörle, A. Türler, M. Wegrzecki, B. Wierczinski, A. Yakushev, and A. Yeremin. Observation of the $3n$ evaporation channel in the complete hot-fusion reaction $^{26}\text{Mg} + ^{248}\text{Cm}$ leading to the new superheavy nuclide ^{271}Hs . *Physical Review Letters*, 100(13):132503, April 2008.
- [10] A. Türler. Nuclear structure and reaction studies near doubly magic ^{270}Hs . *Radiochimica Acta*, 100(2):75–83, February 2012.
- [11] André Von Zweidorf, W. Bröchle, S. Bürger, H. Hummrich, Jens Volker Kratz, B. Kuczewski, G. Langrock, U. Rieth, M. Schädel, N. Trautmann, K. Tsukada, and N. Wiehl. Evidence for the formation of sodium hassate(VIII). *Radiochimica Acta*, 92(12):855–861, December 2004.
- [12] D.N. Poenaru, M. Ivascu, and D. Mazilu. A new semiempirical formula for the alpha decay half-lives. *Journal de Physique Lettres*, 41(24):589–590, 1980.
- [13] E. Rurarz. Alpha emitters in the region from Ce to Os. *Acta Physica Polonica B*, 14(12):917–951, 1983.
- [14] Yu. Ts. Oganessian, V. K. Utyonkov, Yu. V. Lobanov, F. Sh. Abdullin, A. N. Polyakov, I. V. Shirokovsky, Yu. S. Tsyganov, G. G. Gulbekian, S. L. Bogomolov, B. N. Gikal,

- A. N. Mezentsev, S. Iliev, V. G. Subbotin, A. M. Sukhov, A. A. Voinov, G. V. Buklanov, K. Subotic, V. I. Zagrebaev, M. G. Itkis, J. B. Patin, K. J. Moody, J. F. Wild, M. A. Stoyer, N. J. Stoyer, D. A. Shaughnessy, J. M. Kenneally, P. A. Wilk, R. W. Loughheed, R. I. Il'kaev, and S. P. Vesnovskii. Measurements of cross sections and decay properties of the isotopes of elements 112, 114, and 116 produced in the fusion reactions $^{233,238}\text{U}$, ^{242}Pu , and $^{248}\text{Cm} + ^{48}\text{Ca}$. *Physical Review C*, 70(6):064609, December 2004.
- [15] M. Brack, Jens Damgaard, A. S. Jensen, H. C. Pauli, V. M. Strutinsky, and C. Y. Wong. Funny Hills: The Shell-Correction Approach to Nuclear Shell Effects and Its Applications to the Fission Process. *Reviews of Modern Physics*, 44(2):320–405, April 1972.
- [16] F.R. Xu, P.M. Walker, J.A. Sheikh, and R. Wyss. Multi-quasiparticle potential-energy surfaces. *Physics Letters B*, 435(3-4):257–263, September 1998.
- [17] Hai-Yan Meng, Hua-Lei Wang, and Min-Liang Liu. Landscape appreciation of systematic structure properties in even-even nuclei along the valley of stability. *Physical Review C*, 105(1):014329, January 2022.
- [18] William D. Myers and Wladyslaw J. Swiatecki. Nuclear masses and deformations. *Nuclear Physics*, 81(1):1–60, June 1966.
- [19] V.M. Strutinsky. Shell effects in nuclear masses and deformation energies. *Nuclear Physics A*, 95(2):420–442, April 1967.
- [20] S. Cwiok, J. Dudek, W. Nazarewicz, J. Skalski, and T. Werner. Single-particle energies, wave functions, quadrupole moments and g-factors in an axially deformed woods-saxon potential with applications to the two-centre-type nuclear problems. *Computer Physics Communications*, 46(3):379–399, 1987.
- [21] H.C. Pradhan, Y. Nogami, and J. Law. Study of approximations in the nuclear pairing-force problem. *Nuclear Physics A*, 201(2):357–368, 1973.
- [22] P. Möller and J.R. Nix. Nuclear pairing models. *Nuclear Physics A*, 536(1):20–60, 1992.